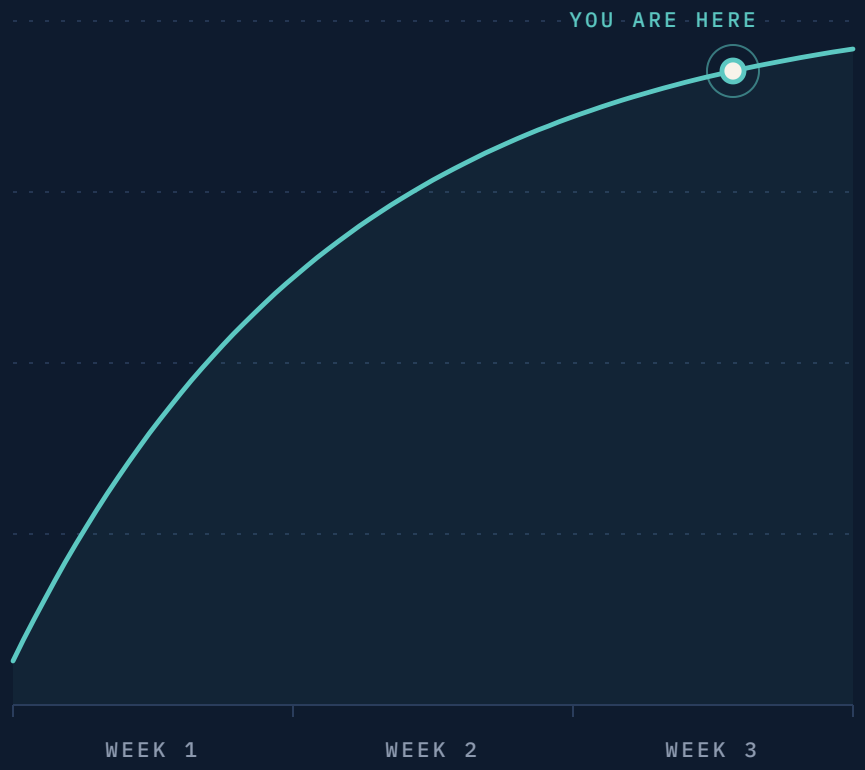
WEEK 3 · DAY 3 · OVERVIEW

Spread Duration and DTS



Created by **Jeff Lenamon**

ABOUT 45 MIN WITH EXERCISES / 8 SECTIONS / 4 EXERCISES

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

The Desk Lead's Note

Two more columns of our month-end file: spread_duration and dts. In every one of the 200 rows spread_duration is the same number as duration. A copy error? Rebuild both columns from the bond math, and tell me how the book's DTS splits by rating bucket.

Your Route Through Day 3

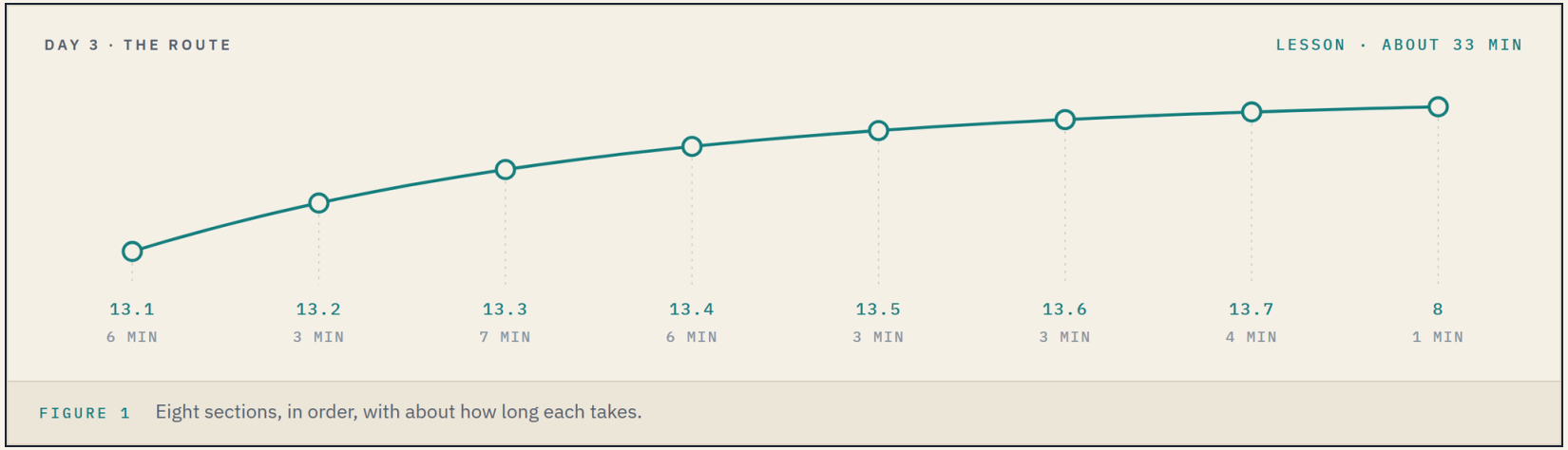


FIGURE 1 Eight sections, in order, with about how long each takes.

	SECTION	ABOUT
13.1	Bump the Spread, Reprice	6 min
13.2	Why It Equals Duration Here	3 min
13.3	`spread_duration`, Docstring First	7 min
13.4	DTS: Spread Duration Times Spread	6 min

	SECTION	ABOUT
13.5	The File's Two Columns, Rebuilt	3 min
13.6	When Did 10000 Arrive?	3 min
13.7	The Book's DTS by Rating Bucket	4 min
8	Bring It Home	1 min

Time. About 33 minutes for the lesson, then about 12 for the four exercises.

13.1 Bump the Spread, Reprice

FROM A SLOPE TO YEARS

```
dirty = bondmath.dirty_price(settlement, maturity, coupon_pct, yield_pct)
spread_duration_years = (price_down - price_up) / (2 * 0.0001) / dirty
print(f'dirty price {dirty:.6f}; spread duration {spread_duration_years:.6f} years')
```

```
dirty price 108.709928; spread duration 4.626971 years
```

EXCEL SIDE BY SIDE

JOB	PYTHON	EXCEL	RESULT
Price at the yield (B1)	`bondmath.price(settlement, maturity, coupon_pct, yield_pct)`	`=PRICE(DATE(2026,9,30),DATE(2032,6,18),7/100,5.606/100,100,2,0)`	106.72659513009698
Spread duration (B5)	`spread_duration_years`	`=(B2-B3)/(2*0.0001)/(B1+B4)`	4.626970799149981

Bezane's price moves about 4.63 percent of its value for 100 bp of spread: spread duration is 4.626971 years.

13.2 Why It Equals Duration Here

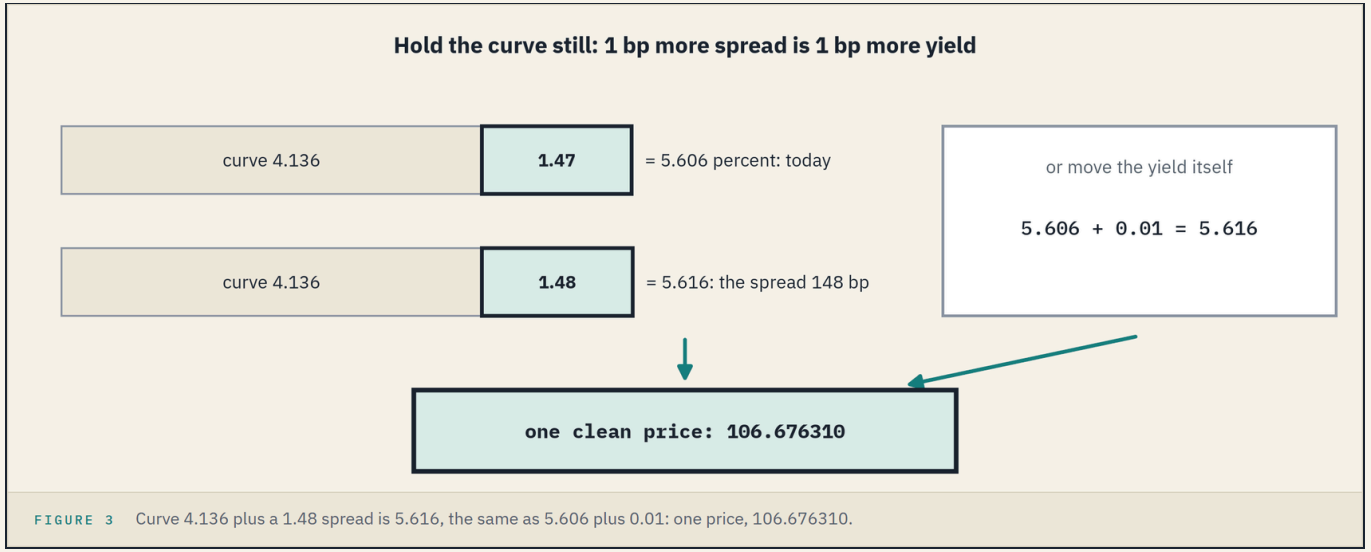


FIGURE 3 Curve 4.136 plus a 1.48 spread is 5.616, the same as 5.606 plus 0.01: one price, 106.676310.

EXCEL SIDE BY SIDE

JOB	PYTHON	EXCEL	RESULT
Against the yield moved	`price_wider - price_up`	`=B11-B3`	0

Notebook 12 found the file's rule: yield = curve yield + spread / 100.

13.3 spread_duration, Docstring First

ONE BOND AGAINST EXCEL

```
bezane_sd = bondmath.spread_duration(settlement, maturity, coupon_pct, yield_pct)
print(f'bondmath {bezane_sd:.12f}, Excel {excel_b5:.12f}')
print(f'gap {abs(bezane_sd - excel_b5):.1e} years')
```

```
bondmath 4.626970799151, Excel 4.626970799150
gap 1.3e-12 years
```

ALL 200 AGAINST EXCEL

```
holdings['spread_duration_bm'] = [bondmath.spread_duration(*bond) for bond in bonds]
print(f'largest gap to Excel over 200 bonds: {gap.max():.1e} years')
```

```
largest gap to Excel over 200 bonds: 3.0e-11 years
```

`spread_duration` **matches Excel's bumped** `PRICE` **on all 200 bonds and sits within 0.00002 years of modified duration.**

13.4 DTS: Spread Duration Times Spread

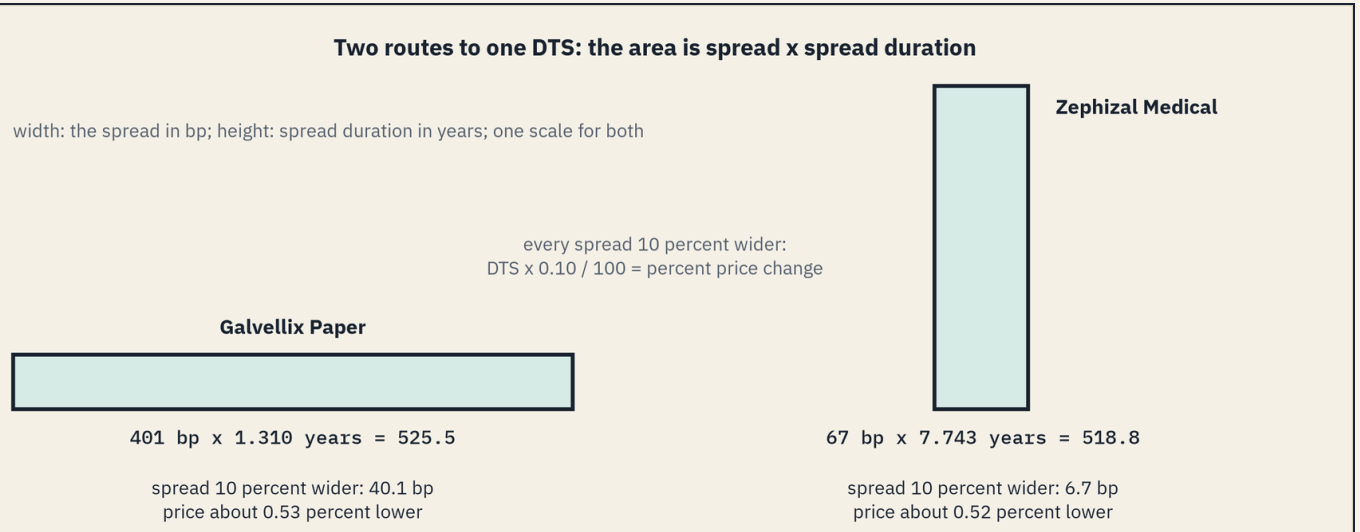


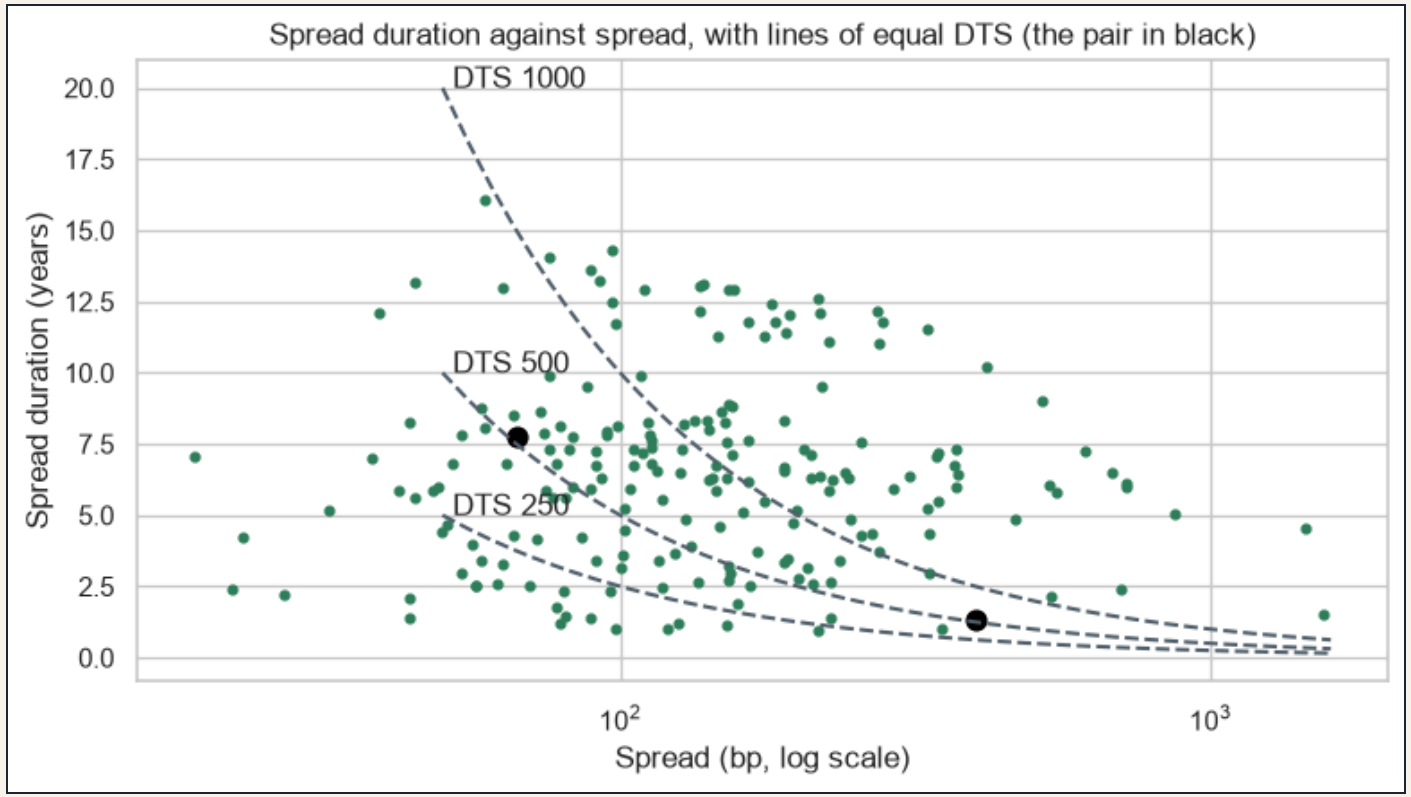
FIGURE 4 Galvellix Paper (401 bp x 1.310) and Zephizal Medical (67 bp x 7.743) reach DTS 525.5 and 518.8.

TWO BONDS, ONE DTS

	ISSUER	OAS	SPREAD_DURATION_BM
71	Galvellix Paper	401	1.310
196	Zephizal Medical	67	7.743

Two bonds with the same DTS lose about the same share of value when every spread moves by the same share, whatever their duration and spread.

Every Bond, With Lines of Equal DTS



BEZANE'S DTS

```
bondmath.dts 680.1647; the file's dts 680.2
```

EXCEL SIDE BY SIDE

JOB	EXCEL	RESULT
DTS (B7)	`=B5*147`	680.1647074750472

A bond on a dashed line has that DTS, however it got there: short and wide, or long and tight.

13.5 The File's Two Columns, Rebuilt

SPREAD_DURATION AGAINST THE FILE

```
print(f'largest gap {gap_sd.max():.9f} years')
```

```
largest gap 0.000498707 years
```

DTS FROM THE FILE'S OWN COLUMNS

```
print(f'largest gap {from_file.max():.13f}')
```

```
largest gap 0.0500000000002
```

DTS FROM THE BOND MATH

```
print(f'largest gap {from_math.max():.4f}; {(from_math > 0.05).sum()} of 200 bonds above 0.05')
```

```
largest gap 0.3999; 82 of 200 bonds above 0.05
```

Both columns rebuilt: what differs is the file's rounding, never the math.

13.6 When Did 10000 Arrive?

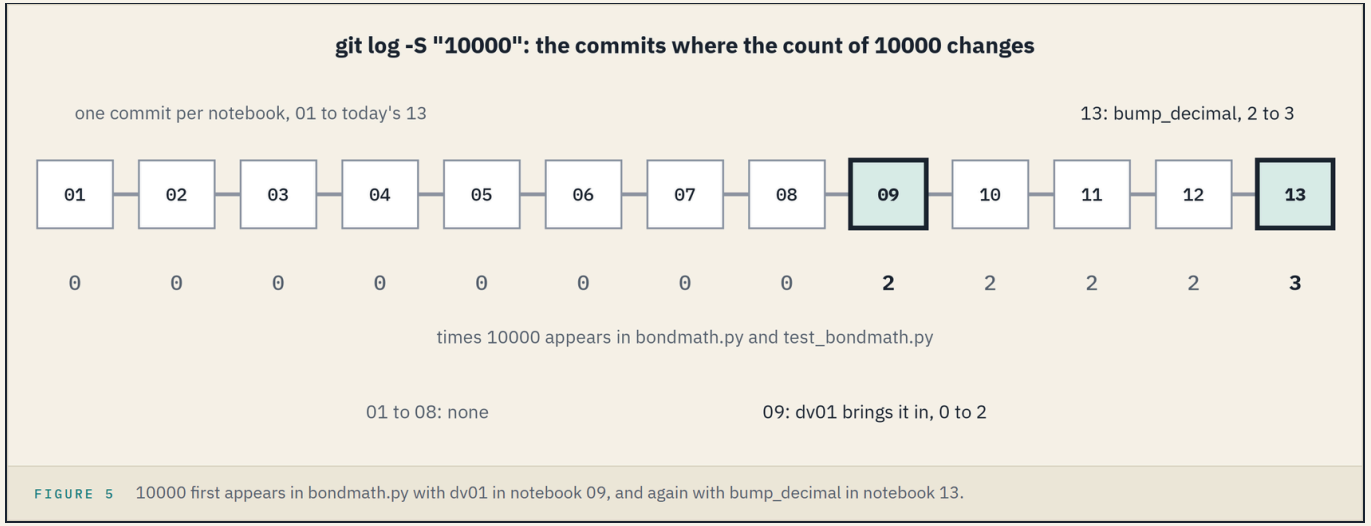


FIGURE 5 10000 first appears in bondmath.py with dv01 in notebook 09, and again with bump_decimal in notebook 13.

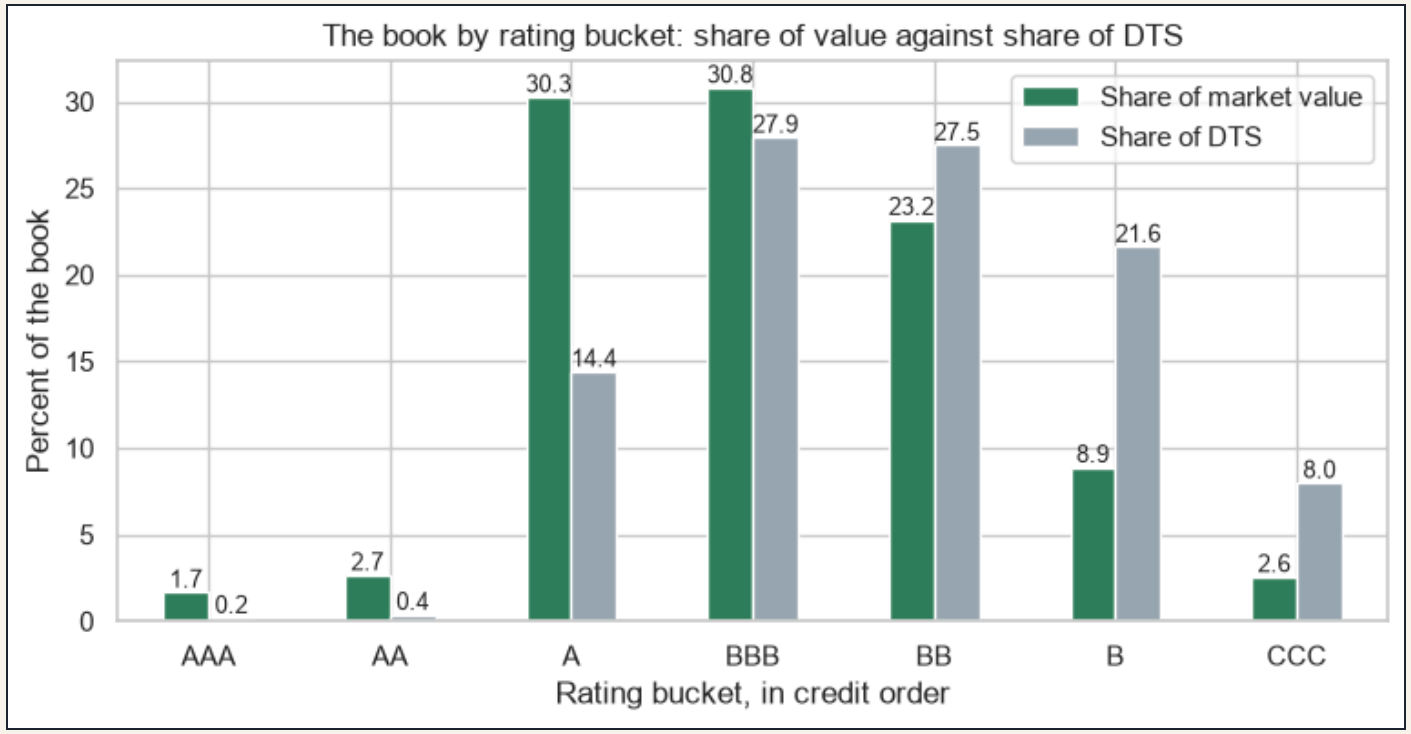
THE PICKAXE

```
!git -C "{work}" log -S "10000" --oneline
```

```
e42ff20 Notebook 13: spread_duration and dts
5fcb7c6 Notebook 09: macaulay_duration, modified_duration, and dv01
```

> `git log -S` **finds the commit that brought a number in: 10000 arrived with** `dv01` **in notebook 09.**

13.7 The Book's DTS by Rating Bucket



BY GRADE

GRADE	WEIGHT_PCT	SHARE_OF_DTS_PCT
High Yield	34.6	57.1
Investment Grade	65.4	42.9

EXCEL SIDE BY SIDE

`=SUMIFS(O2:O201,B2:B201,"BBB")` returns 297.2402393040499, BBB's row; every bucket's `SUMIFS` equals the table to 1e-9.

High Yield is 34.6 percent of the book's value and 57.1 percent of its DTS: the dollars and the spread risk sit in different buckets.

Bring It Home

IN POWERSHELL

```
Set-Location "$HOME\projects\my_bondmath"
$work = "$env:TEMP\pfi_course2_13_k4w7q2xr"
Copy-Item "$work\bondmath.py", "$work\test_bondmath.py", "$work\.gitignore" .
python -m pytest -q
git diff --stat
git add bondmath.py test_bondmath.py
git commit -m "Notebook 13: spread_duration and dts" -m "Spread duration and DTS feed the risk report of notebook 14"
git log -S "10000" --oneline
git log -S "10000" --oneline -- test_bondmath.py
```

The copy replaces your files: anything of your own must be in the work folder first.

You Can Now

- Measure spread duration by moving the spread 1 bp each way and repricing, with **`spread_duration`** or three **`PRICE`** cells.
- Say why, for a bond priced at the curve plus a spread, spread duration is modified duration, and why the bump sits a hair above it.
- Compute DTS with **`dts`** and say what it measures: the price change when a spread moves by a share of itself.
- Rebuild the file's **`spread_duration`** and **`dts`** columns, and name the rounding that separates them.
- Add up the book's DTS by bond and by rating bucket, with pandas or with **`SUMPRODUCT`** and **`SUMIFS`**.
- Find the commit that brought a number in, with **`git log -S`**.

NEXT
WEEK 3 · DAY 4

Portfolio Aggregation and Scenarios

The 200 bonds rolled into one book, its risk card, and the book repriced in full under rate and spread what-ifs.